Resilience Academy

SDI Training for ICT Professionals

CRD Platform Walkthrough

Exploring Tanzania's Climate Risk Database

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Day 2 | 1 Hour (Lecture & Demo) | crd.resilienceacademy.ac.tz

Resilience Academy

Session Overview

What we will cover:

1. What is CRD and why does it matter?
2. Navigating the catalogue — browsing, searching, filtering
3. Reading a dataset page — metadata, maps, downloads
4. Understanding WMS/WFS links (the "share" buttons of geodata)
5. Quick look at the admin panel
6. Hands-on: find datasets for Tanga and Mwanza

Browse the Catalogue
↓
Read Metadata
↓
Preview on Map
↓
Download / Connect

By the end of this session

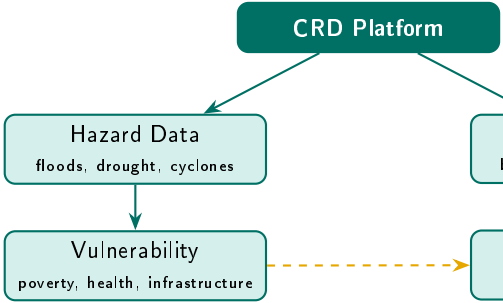
You will know how to find, preview and download

What is CRD?

Climate Risk Database (CRD) is an online platform where climate, hazard, exposure, and vulnerability data for Tanzania are stored, shared, and explored.

Think of it like a library, but for maps and data:

- A **physical library** has shelves, a catalogue, and a librarian
- CRD has **categories**, a **search bar**, and an **admin panel**
- Instead of borrowing books, you **download datasets** or **connect directly** from GIS software



Navigating the Catalogue: Browse, Search, Filter

Three ways to find data:

1. **Browse by Category**
   - Click **"Data"** in the top menu
   - Categories appear on the left sidebar
   - Examples: *Hazard, Exposure, Vulnerability, Baseline*
2. **Search by Keyword**
   - Type in the search bar: e.g. `flood Tanga`
   - Results update instantly as you type
   - Works with dataset names, descriptions, and keywords

CRD Top Menu Bar

↓

Data — All datasets, layers, documents

↓

Maps — Pre-made map compositions

↓

About — Platform information

Search bar: `flood Tanga`

↓

Results: 12 datasets found

Flood hazard map, Flood risk zones...

The Dataset Page: Everything About One Layer

When you click on a dataset, you see a **detail page** with several sections:

Left side — Map Preview

- Interactive map showing the data
- Zoom in/out, pan, click features for info
- Legend showing colours and categories

Right side — Metadata

- **Title** and **Abstract**: what the data is about
- **Keywords**: searchable tags (e.g. *flood, hazard, Tanga*)
- **Category**: which group it belongs to

Bottom section — Actions

- **Download**: get the actual data files
  - Shapefile (.shp) — for GIS software
  - GeoJSON — for web applications
  - CSV — for spreadsheets
  - GeoTIFF — for raster/image data
- **Metadata**: view full ISO metadata record
- **Map**: open in the built-in map viewer
- **Share**: get a link to share with colleagues

Developer Page

Click **"API"** to see the technical endpoints

WMS/WFS Endpoints — The "Share Links" of Geodata

Analogy: Think of YouTube videos.

- You can **download** a video (like downloading a shapefile)
- Or you can **stream** it via a link (like WMS/WFS)

WMS — Web Map Service

- Sends you a **picture** of the map
- Like a screenshot — you can look but not edit
- Great for **background layers** in QGIS

WFS — Web Feature Service

- Sends you the **actual data** (points, lines,

What the URLs look like:

WMS endpoint

```
https://crd.resilienceacademy.ac.tz/
geoserver/ows?service=WMS&
version=1.1.1&request=GetMap&
layers=geonode:flood_tanga
```

WFS endpoint

```
https://crd.resilienceacademy.ac.tz/
geoserver/ows?service=WFS&
version=1.0.0&request=GetFeature&
typeName=geonode:flood_tanga
```

WMS	WFS
Returns: Image (PNG/JPEG)	Returns: Data (GML/JSON)

Admin Panel: What Administrators Can Do

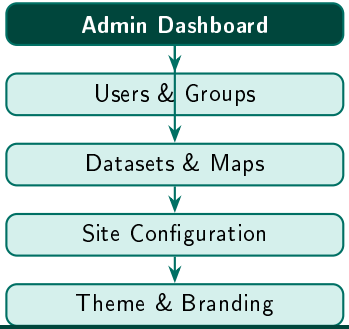
The **Admin Panel** is the control room of CRD. Only users with admin rights can access it.

What you can manage:

- **Users & Groups**: create accounts, assign roles (viewer, editor, manager, admin)
- **Datasets**: upload, edit metadata, set permissions, delete
- **Maps**: create and manage map compositions
- **Site Settings**: change name, logo, home page text

Access the admin panel:

1. Log in with an admin account
2. Click your username (top-right corner)
3. Select **"Admin"** from the dropdown



Exercise: Explore CRD Together

Open your browser and go to:
https://crd.resilienceacademy.ac.tz

Task 1: Find Tanga Data

1. Click **"Data"** in the menu
2. In the search bar, type **Tanga**
3. How many datasets appear?
4. Click on one — read the abstract
5. Find the **download button** — what formats are available?

Task 2: Find Mwanza Data

1. Clear the search and type **Mwanza**
2. Filter by **category** — try "Hazard"

Task 3: Find the WMS Link

1. On any dataset page, look for **"API"** or **"Developer"** section
2. Copy the WMS link — you will use it on Day 4
3. Paste it in a text file or note for later

Discussion Questions:

- What types of data did you find?
- Was the metadata complete and clear?
- What data is missing that your institution could contribute?

Discussion: How Would CRD Help YOUR Institution?

Think about your daily work:

Data Sharing

Do you currently share geographic data with other departments? How? (USB drives? Email? WhatsApp?)

CRD provides a **single, organised location** accessible from any browser.

Data Discovery

When you need a map or dataset, how do you find it? Do you know what data other departments have?

Data Preservation

What happens when a staff member leaves? Are their datasets backed up?

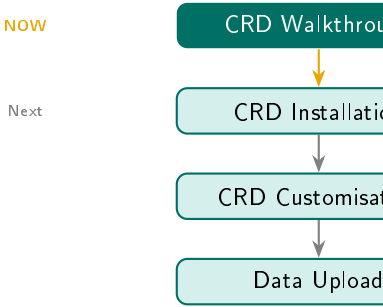
CRD ensures data **stays in the institution**, not on personal laptops.

Collaboration

Could teams in Tanga and Mwanza work on the same data without sending files back and forth?

CRD enables **real-time access** from any location.

Session Summary

What we covered:

- ✓ CRD is Tanzania's **Climate Risk Database** — built on GeoNode
- ✓ You can **browse, search, and filter** datasets through a web browser
- ✓ Each dataset has **metadata, a map preview, and download options**
- ✓ **WMS** streams map images; **WFS** streams actual data
- ✓ The **admin panel** lets administrators manage users, data, and settings
- ✓ You explored Tanga and Mwanza datasets hands-on